2. Requirement Analysis Phase

2.1 Introduction

The Requirement Analysis Phase identifies, analyzes, and documents the requirements of EduGenie: Google Gemini Powered Learning Assistant. It focuses on understanding the problems faced by learners and defining the features needed to solve them. EduGenie is designed as an intelligent learning assistance system that uses Google Gemini to explain concepts, answer doubts, generate practice material, and help learners plan their studies.

2.2 Problem Analysis

Self-study can be difficult. Learners may not understand a concept from textbooks, may not have immediate access to a teacher, may find it hard to revise large amounts of material, and may not know how to test their understanding. Traditional resources are generally static and do not adapt to the individual learner. The major problems identified are:

- Difficulty in understanding complex concepts quickly.
- Lack of immediate, personalized doubt resolution.
- Time-consuming manual preparation of summaries and notes.
- Lack of practice questions tailored to a specific topic.
- Poor study planning and time management.
- Difficulty in tracking learning progress.
- Information overload from generic search results.

2.3 Objectives of Requirement Analysis

- To identify the needs and expectations of learners.
- To define the major functionalities of EduGenie.
- To identify the required hardware and software resources.
- To determine functional and non-functional requirements.
- To define requirements for Gemini-based AI integration.
- To provide a clear foundation for design and development.

2.4 Functional Requirements

2.4.1 User Registration and Authentication

- User registration and secure login.
- User profile management.
- Logout functionality.
- Basic authentication and input validation.

2.4.2 AI Tutor Chat

- Accept questions in natural language.
- Generate clear, structured answers using Gemini.
- Maintain conversation context for follow-up questions.
- Allow the learner to choose the subject and difficulty level.

2.4.3 Concept Explanation

- Explain a topic at beginner, intermediate, or advanced level.
- Provide examples, analogies, and key points.

2.4.4 Notes Summarization

- Accept pasted text or notes as input.
- Generate a concise summary and key takeaways.
- Allow the summary to be saved for revision.

2.4.5 Quiz Generation and Evaluation

- Generate quizzes on a chosen topic, difficulty, and number of questions.
- Evaluate the learner's answers and calculate the score.
- Provide correct answers with short explanations.

2.4.6 Flashcards

- Generate flashcards from a topic or from supplied notes.
- Allow learners to flip and review cards.

2.4.7 Study Plan Generator

- Accept subject, topics, available time, and target date.
- Generate a structured day-wise or week-wise study plan.

2.4.8 History and Dashboard

- Store previous chats, summaries, quizzes, and plans.
- Display quiz scores, topics studied, and activity trends through charts.

2.5 Non-Functional Requirements

Requirement	Description
Usability	Simple, clean, and responsive interface usable without technical knowledge.
Performance	AI responses and page loads should appear within a reasonable time.
Security	Secure authentication, protected APIs, and safe storage of the Gemini API key on the server.
Reliability	Accurate storage of data and graceful handling of API errors or timeouts.
Scalability	Architecture should allow new subjects, features, and users to be added.
Maintainability	Modular, well-organized code that is easy to test and extend.

2.6 Hardware Requirements

- Computer or laptop with minimum 4 GB RAM and 10 GB free storage.
- Internet connection (required for Gemini API access).
- Keyboard and mouse; smartphone or computer for accessing the application.

2.7 Software Requirements

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python, FastAPI
Database	SQLite / PostgreSQL
AI / LLM	Google Gemini API
Authentication	JWT
IDE	Visual Studio Code
Version Control	Git and GitHub
Operating System	Windows / Linux / macOS

2.8 Expected Outcome

The phase produces a well-defined set of requirements that serves as the foundation for the planning, design, development, testing, and demonstration phases. EduGenie is expected to give learners a convenient platform for understanding concepts, practicing, and organizing their studies with AI support.